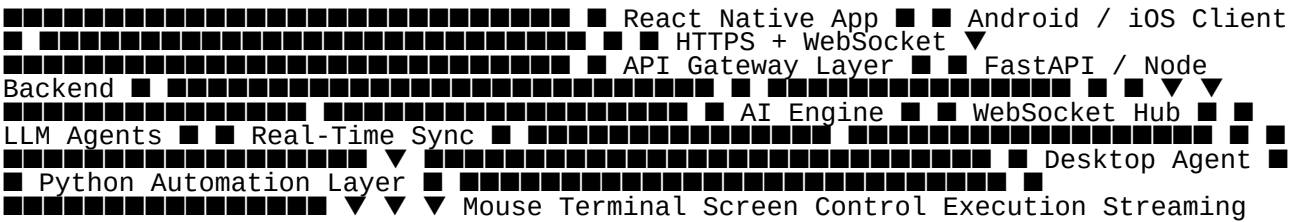


AI Remote Control System

Complete Data Flow & Architecture Diagrams

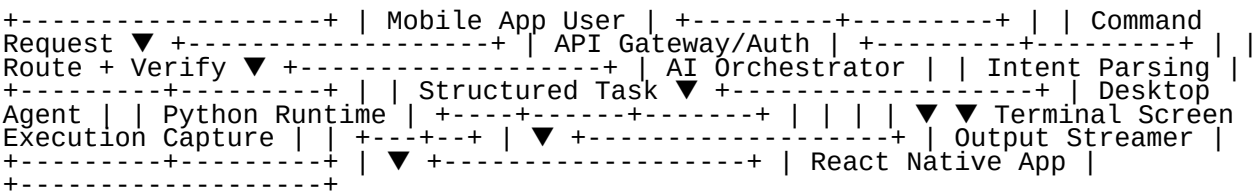
This document contains detailed architectural diagrams, system data flows, AI orchestration layers, communication pipelines, and deployment structures for the AI-powered remote desktop and automation platform.

1. High-Level System Architecture



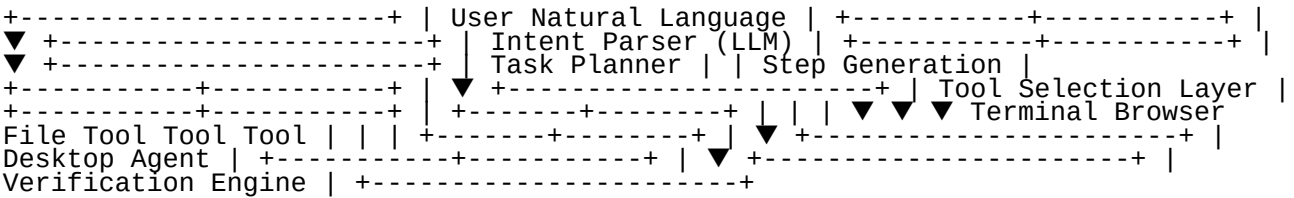
2. Complete Data Flow Diagram

User Action Flow: 1. User taps action on mobile 2. Request sent via WebSocket/API 3. Backend validates token 4. AI engine processes instruction 5. Command routed to desktop agent 6. Desktop executes task 7. Output streamed back



3. AI Integration Architecture

The AI layer acts as the autonomous decision-making engine for the system.

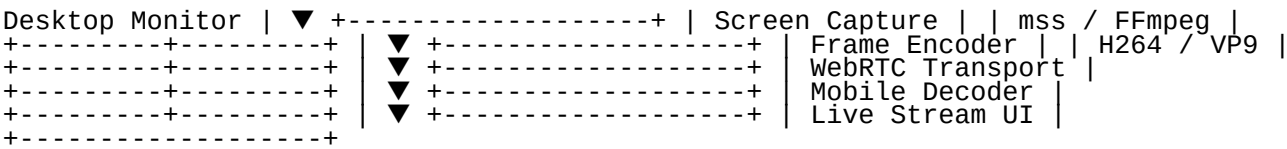


4. WebSocket Communication Flow

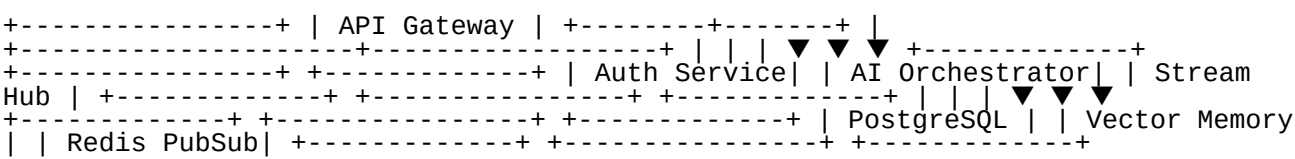
Real-time communication powers: • Terminal outputs • Screen streaming • AI responses • Live command execution



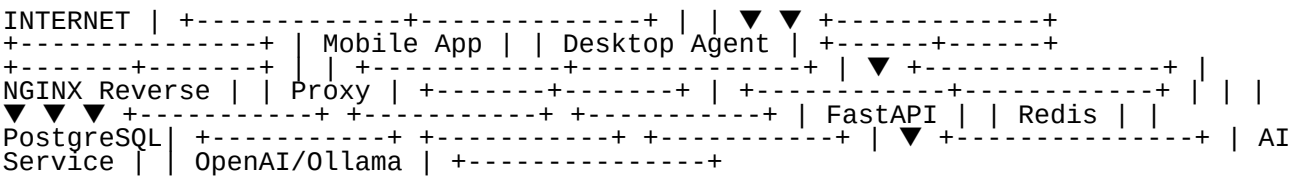
5. Screen Streaming Pipeline



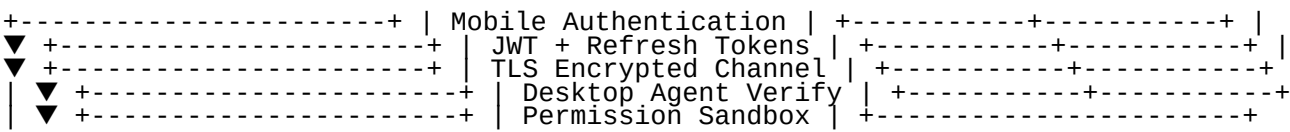
6. Backend Microservice Architecture



7. Deployment Architecture



8. Security Architecture



9. Recommended Future Scaling

Scaling Roadmap: • Single-user local architecture • Multi-device synchronization • Cloud orchestration • Distributed AI agents • Enterprise dashboard • Kubernetes deployment • Multi-region streaming • Autonomous workflow marketplace

Final Note:

This architecture combines concepts from remote desktop systems, AI agents, cloud-native systems, operating system automation, and distributed real-time communication. It is suitable for advanced portfolio development, startup MVPs, and enterprise-grade automation platforms.